

From Theory to Applications in Knowledge Compilation and Model Counting

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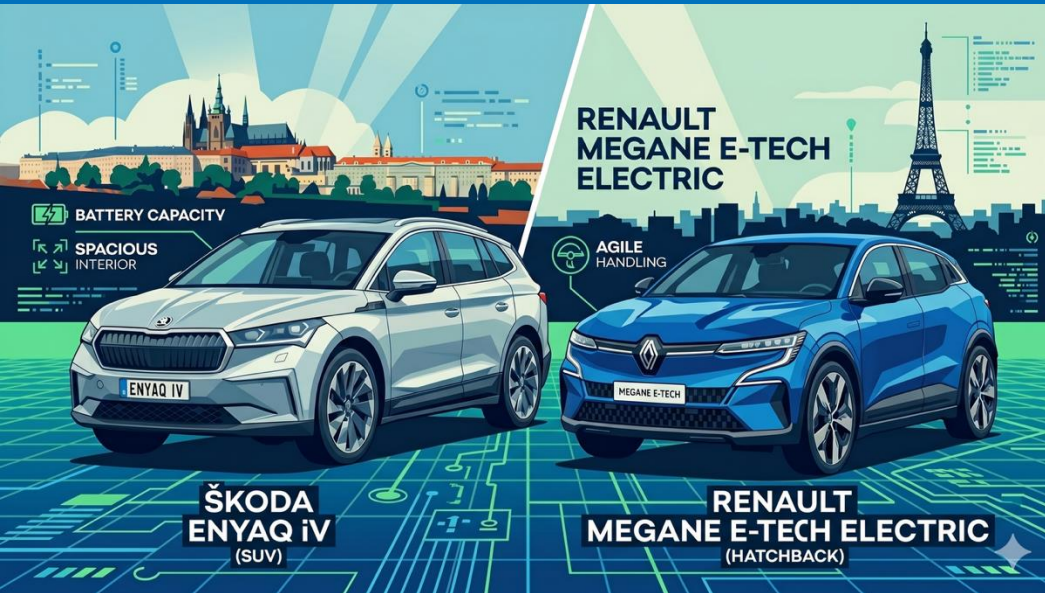
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Automotive Industry

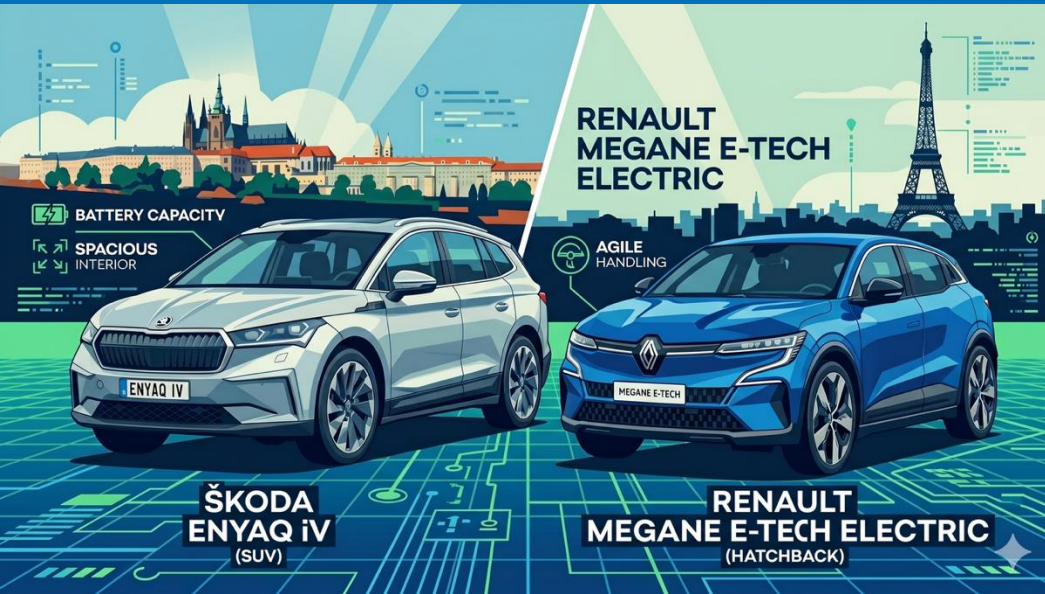


Automotive Industry



thousands of variables, millions of rules
engine, transmission, electronics, trim, ...

Automotive Industry



Client configuration scenario

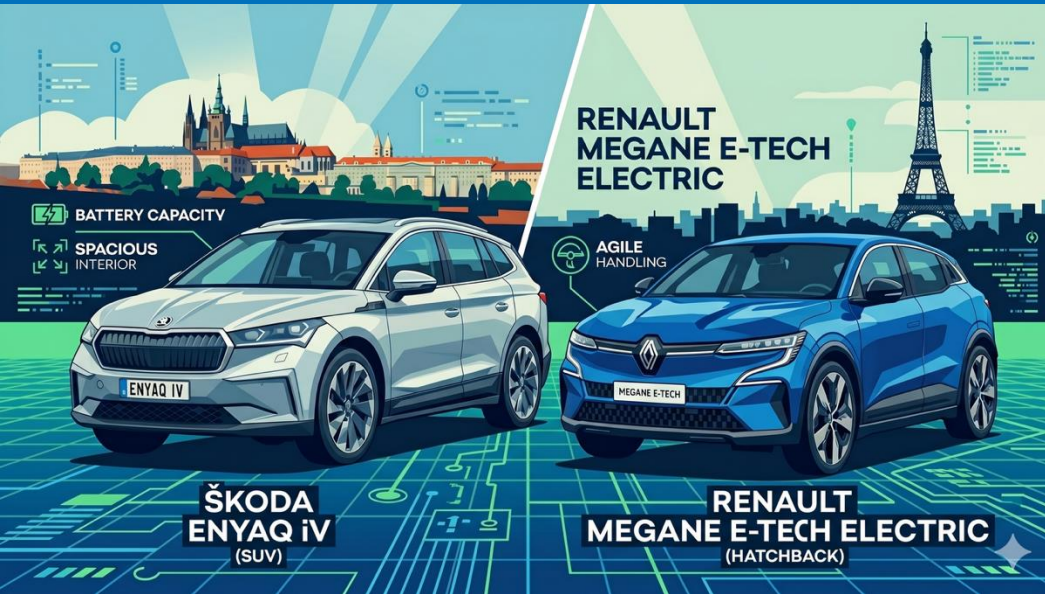
Can the car actually be built?

Is this combination of options valid?

thousands of variables, millions of rules

engine, transmission, electronics, trim, ...

Automotive Industry



thousands of variables, millions of rules
engine, transmission, electronics, trim, ...

Client configuration scenario

Can the car actually be built?
Is this combination of options valid?

Engineering scenario

Which combinations are possible?
Which combinations must the
subsystem support?

Automotive Industry: Knowledge Compilation



rules

logical formulae



Automotive Industry: Knowledge Compilation



rules

logical formulae



compilation

slow, but done once



tractable representation

many operations in linear time



operations



Automotive Industry: Knowledge Compilation



rules

logical formulae

open challenge

d4, c2d, ...

compilation

slow, but done once

DNNF circuit

strictly more compact

decision-DNNF circuit

tractable representation

many operations in linear time

operations



Automotive Industry: Knowledge Compilation



rules

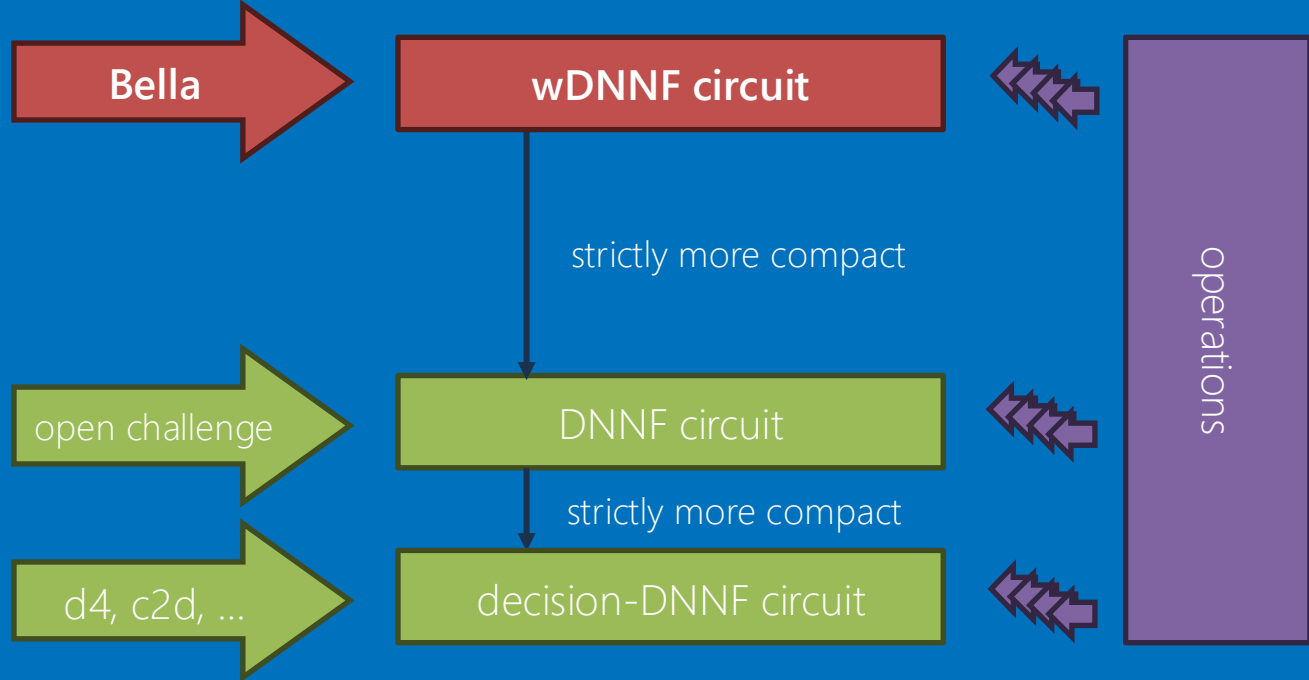
logical formulae

compilation

slow, but done once

tractable representation

many operations in linear time



Automotive Industry: Experimental Results

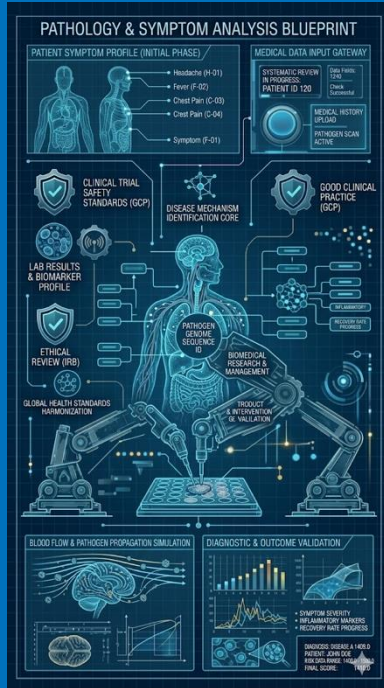
Configuration instance	wDNNF circuit		decision-DNNF circuit		improvements	
	Bella		d4		speed-up factor (×)	size reduction (%)
	time (s)	size	time (s)	size		
C202_FS	66.3	4 245 485	1 256.2	154 862 147	18.9×	97.26%
C202_FW	70.7	7 344 961	---*	---*	N/A	N/A
C210_FS	33.2	4 300 692	1 065.3	240 087 456	32.1×	98.21%
C210_FW	56.7	6 429 027	4 048.2	715 624 258	71.4×	99.10%

Compilation times (in seconds) and circuit sizes for the four **most challenging small-scale real-world configuration instances**.

* timeout after 7200s

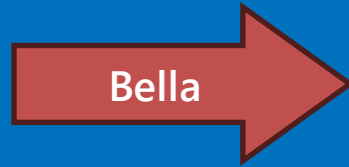


Medical Diagnostics: Knowledge Compilation



rules

logical formulae



compilation

slow, but done once



strictly more compact



strictly more compact



strictly more compact



tractable representation

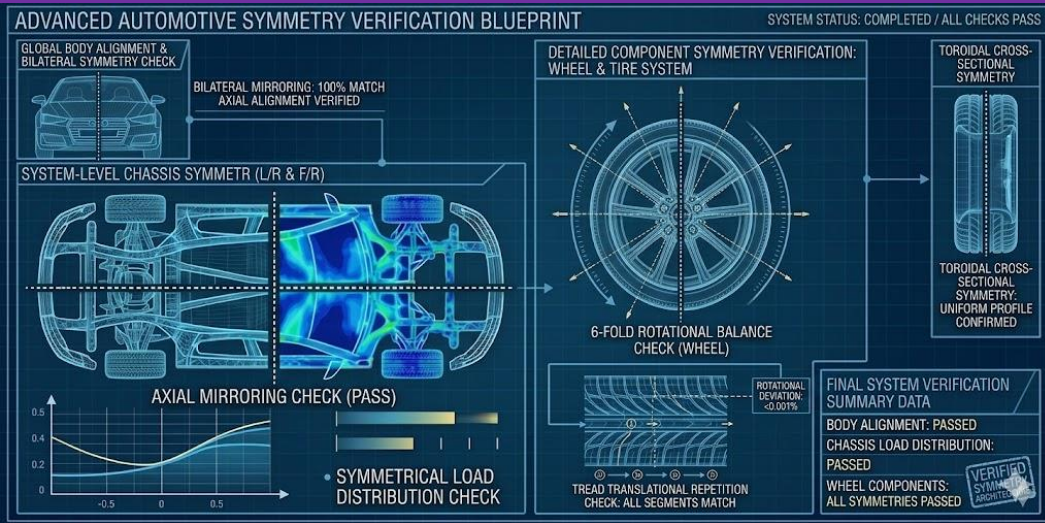
many operations in linear time



operations



Automotive Industry: Model Counting

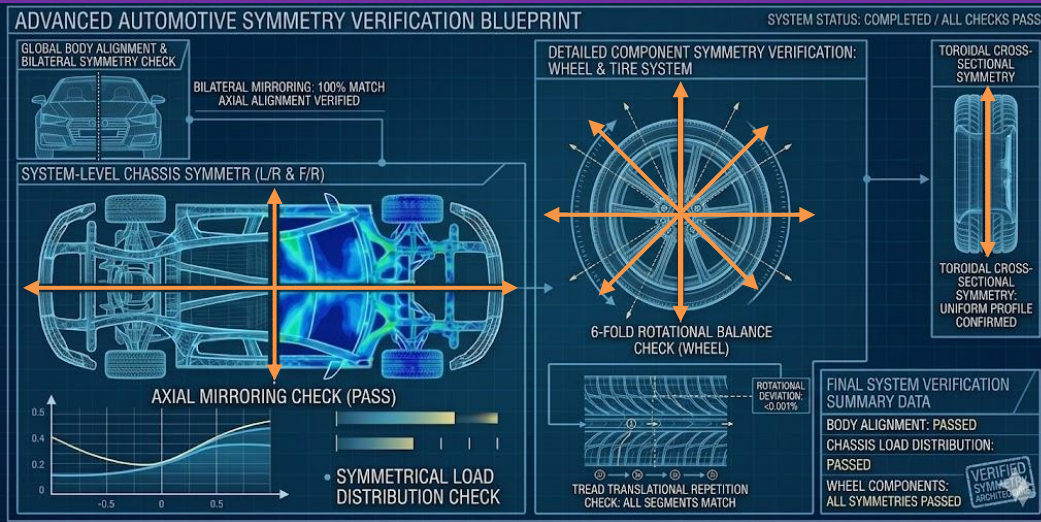


Counting question

How many valid combinations are possible?

Computationally harder than the previous scenarios

Automotive Industry: Model Counting



Counting question

How many valid combinations are possible?

Symmetries

Many combinations are equivalent under certain transformations

Computationally harder than the previous scenarios

We can exploit symmetries to make it easier...

But detecting symmetries is hard...

Cara vs SymGanak: Experimental Results

Highly symmetric instances	Cara	SymGanak
Battleship problems	1	0
ProbLog programs	6	5
N-Queens problem	13	12
Latin square problems	20	17
Counting principle	3	6
k-coloring graph problems	21	18
Parity principle	8	7
Tseitin transformation problems	7	13
FPGA Switch-Box	11	6
Total	90	84

Number of solved highly symmetric instances.



Cara vs SymGanak: Experimental Results

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Number of solved highly symmetric instances.

Less symmetric instances	Total
Cara	595
SymGanak	512
D4	594

Number of solved less symmetric instances.

Cara performs strongly on both highly symmetric and less symmetric instances.



Model Counting Competition 2025

Place	Model counter		Author(s)	(#)	Solved
1	Ganak		Mate Soos, Kuldeep S. Meel	(2)	60
2	Cara ^{***}		Petr Illner	(1)	50
	D4		Jean-Marie Lagniez, Pierre Marquis	(2)	50
4	SharpSAT-TD-CH		Yipei Deng, Junping Zhou, Jiaxin Liang, Le Xin	(4)	47
	SharpSAT-TD		Tuukka Korhonen, Matti Järvisalo	(2)	47
6	MTMC		Ivor Spence	(1)	39
7	GPMC		Kenta Ito, Kenji Hashimoto	(2)	38

* received the “**Notable Mention: Best Newcomer**” award

** **first-ever symmetry-aware model counter** to participate in the competition



Conference papers:

- Illner, Petr; Kučera, Petr. A Compiler for Weak Decomposable Negation Normal Form. *Proceedings of the AAAI Conference on Artificial Intelligence*. [**AAAI - CORE A***, **lead author**, **poster**]
- Illner, Petr. New Compilation Languages Based on Restricted Weak Decomposability. *Proceedings of the AAAI Conference on Artificial Intelligence*. [**AAAI - CORE A***, **sole author**, **poster**, selected for **oral presentation** (~20% of accepted papers)]

Workshops:

- Illner, Petr. Cara: An Isomorphism-Based #SAT Solver. *Workshop on Counting, Sampling, and Synthesis 2025*. [**sole author**, **oral presentation**]

Competitions:

- Illner, Petr. Cara. *Model Counting Competition 2025*. Second place in Track 1: Model Counting, Ranking A (exact). [**sole author**]

State-of-the-art software:

- Illner, Petr. Bella: A knowledge compiler for wDNNF, pwDNNF, nwDNNF and decision-DNNF circuits. [**sole developer**]
- Illner, Petr. Cara: An isomorphism-aware #SAT solver. [**sole developer**]

